

Evolution of the IEEE P7009 Standard: Towards Fail-Safe Design of Autonomous Systems

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Abstract—Autonomous systems are complex, diverse, and likely be used in environments that change unpredictably. We must develop mechanisms to cope with failures, potentially caused by both the natural stresses of the operating environment and the system's inadequate environmental models. Failures also arise from system component wear. In 2016, the IEEE launched the Global Initiative on Ethics of Autonomous and Intelligent Systems in response to the proliferation of increasingly-autonomous systems. A cornerstone of this initiative is the IEEE P7000 series of projects, which set the standards for the future of ethical intelligent and autonomous technologies. This paper describes the ongoing development of the IEEE P7009 standard on “Fail-Safe Design of Autonomous and Semi-Autonomous Systems”. We provide a collective perspective on the ongoing development of IEEE P7009, and outline the objectives, current approaches, issues under discussion, and two exemplar use cases from different sectors: one high-regulation, the other low-regulation.

Index Terms—Autonomous Systems, Fail Safe, Standards

I. INTRODUCTION

Enabling a system to operate autonomously brings both benefits and challenges. The flexibility of Autonomous and Intelligent Systems (AIS) means that they are likely to be used in environments that will change in ways that cannot be anticipated. The complex environments in which AIS may be used will cause failures, whether due to the stressful nature of the operational environment; or because the internal environmental models that the AIS uses are either wrong, incomplete or, more likely, both. These considerations emphasise the need for *fail-safe* AIS design.

This paper describes the development of the IEEE P7009 standard “Fail-Safe Design of Autonomous and Semi-Autonomous Systems”¹. The standard is intended to provide a basis for developers, acquirers², suppliers, regulators and other

interested stakeholders, to design fail-safe AIS in a robust, transparent, and accountable way. It is being developed by a group of international volunteers who are collaborating in an open, transparent and evidence-informed fashion.

IEEE P7009 is one of 13 standards in the IEEE P7000 series, which set the benchmark for the future of ethical intelligent and autonomous technologies. The series is part of the IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems, whose mission is to:

*“ensure every stakeholder involved in the design and development of autonomous and intelligent systems is educated, trained, and empowered to prioritise ethical considerations so that these technologies are advanced for the benefit of humanity.”*³

The vision of the initiative is captured in *Ethically Aligned Design* [1], which provides a set of recommendations to guide the development and use of ethical AIS. The IEEE Ethics Certification Program for Autonomous and Intelligent Systems (ECPAIS)⁴ is developing certification of ethical properties for AIS, including: Transparency, Accountability, Freedom from Unacceptable Bias (Fairness), Privacy, Responsible Governance as well as Safety and Security. In this respect, Fail Safety is an ethical property of AIS.

The collective view presented in this paper provides insights into the ongoing work of the IEEE P7009 Working Group (WG). We highlight particularly challenging topics and describe, for each, the consensus positions reached by the WG. These positions support the standard’s purpose: to provide a practical, technical baseline of specific methods and tools for designing fail-safe AIS. Note that the final content of P7009 will only be publicly available once the draft standard has been formally published by the IEEE Standard Association.

This paper is organised as follows. §II introduces the IEEE P7009 standard. §III presents some challenging topics that the WG has discussed. §IV describes two ongoing challenges that

The authors thank all participants in, and contributors to, the IEEE P7009 Working Group for their efforts to help develop the standard.

¹<https://standards.ieee.org/project/7009.html>

²A stakeholder who acquires the system from a supplier.

³standards.ieee.org/industry-connections/ec/autonomous-systems.html

⁴<https://standards.ieee.org/industry-connections/ecpais.html>

the WG is working to resolve: how to verify the properties described in IEEE P7009, and methods to design fail-safety into an AIS. §V highlights two exemplar use cases of the IEEE P7009 standard. Finally, §VI concludes with the current status and intended timeline for IEEE P7009.

II. INTRODUCING P7009

Established approaches to safety engineering and the assurance of safety-critical systems have led to a suite of systems- and software-engineering standards, such as IEC 61508/61511, ISO/IEC/IEEE 15288, ISO/IEC/IEEE 12207 and sector specific standards such as DO-178C (aviation), ISO26262 (automotive), IEC 62034 (medical), Def Stan 00-056, MIL-STD-882 (defence) and various European norms e.g., EN50128, EN50129 (railway), etc. The majority of these are based on the concept that safety can be engineered into a system through strict observance of normative activities during the product/service life cycle [2].

AIS are less amenable to traditional safety assurance approaches, which rest on the rigour and integrity of the development process. This is because of the inherent uncertainty of an AIS's behaviour as it responds and adapts to its changing environment. We suggest that these challenges make it unlikely that existing fail-safety standards are appropriate for AIS. Assuring the safety of AIS necessitates new approaches [3], and this challenge has motivated the development of the IEEE P7009 standard as encapsulated in the Project Authorisation Request (PAR)⁵:

"This standard establishes a practical, technical baseline of specific methodologies and tools for the development, implementation, and use of effective fail-safe mechanisms in autonomous and semi-autonomous systems. The standard includes (but is not limited to): clear procedures for measuring, testing, and certifying a system's ability to fail safely on a scale from weak to strong, and instructions for improvement in the case of unsatisfactory performance. The standard serves as the basis for developers, acquirers, as well as users and regulators, to design fail-safe mechanisms in a robust, transparent, and accountable manner."

IEEE P7009 contrasts with the legacy systems safety standards, which are focused on safe functional design and operation. The purpose of IEEE P7009 is to identify:

- fail-safe methods to prevent an AIS from causing harm,
- practical means to include these mechanisms in an AIS,
- practical means to characterise the performance of these mechanisms in the operational context of AIS.

Established in 2017, the IEEE P7009 WG adopted the Precautionary Principle [4]: given the intent of the standard, this decision was relatively simple. IEEE P7009 is a foundational standard that remains domain- and application-agnostic. To decide what is or is not safe, and so what is or is not fail-safe, requires knowledge of the system's context and about the consequences of its actions or of inaction. Adding the details of context and consequences to the standard would bind it to specific domains or application areas. Instead, IEEE P7009 offers designers and developers a framework for deciding

⁵A PAR defines the scope, purpose, and contact points for an IEEE Standards Association project.

what is or is not safe or fail-safe for their own domain or application area; because designers and developers are best placed to understand the context and consequences inherent to their system.

The WG has pragmatically adopted the widely accepted adage that a system's safety needs and implies its security. These considerations relate to undesirable change in the AIS with the potential to adversely impact its ability to fail safely, irrespective of the source of the change. Such change will manifest itself as failures and can be captured as part of the wider treatment of failure embodied within the standard.

III. SETTLED CHALLENGES

This section presents some of the challenging topics that have been discussed, and the consensus that has been reached by the IEEE P7009 WG for each topic. This provides an insight into the difficulties that arise when drafting a standard, not least in providing usable definitions and approaches to what are often ill-defined or nebulous concepts.

A. Descriptions, Not Definitions

Unsurprisingly, the topic of definitions has been a source of debate and discussion. In the end, the WG has opted to avoid the use of definitions where possible. The standard compensates for the lack of definitions with an approach that is similar to that used by ISO26262, where terms are characterised by descriptions without being prescriptive.

The use of descriptions enables adopters of the standard to employ the definition that is most appropriate for their sector or application. The descriptions typically reflect both experience and established practice in a given context. Whereas, if IEEE P7009 used definitions, it would run the risk of being ignored or not being adopted, if they were not precisely aligned with existing terminology. By placing the explicit responsibility upon adopters to arrive at specific and agreed-upon interpretations of the descriptive characteristics, we reduce the risk of misinterpretation and consequential adverse outcomes.

B. What is Autonomy?

Given the standard's aims it is no surprise that the question of what constitutes 'autonomy' has also exercised the WG. Here too, the use of descriptions rather than definitions helps to avoid the consequences of adopting an unsuitable definition. With too broad a definition, nearly every computer-based system can be considered autonomous. With a definition too narrow, the scope of the standard becomes too exclusive.

Broadly, we describe an AIS as one that can make decisions without human intervention. Key distinguishing characteristics of an AIS include the intrinsic capabilities to:

- make decisions without any requirement for prior human validation of the consequences of these decisions,
- act upon these decisions through interaction with the real (physical) world, and
- adapt and modify both the decision making process itself and the consequential interactions with the real world through a process of learning.

The standard is constructed to enable stakeholders to determine whether a system is sufficiently autonomous to merit its use. Adopters are best-placed to make this choice for their particular system, thus *autonomy is in the eye of the beholder*.

Similarly, definitions exist that distinguish between full- and semi-autonomy (e.g. [5]). However, the standard treats autonomy as a continuum, placing the same requirements on systems that are autonomous as it does on those that are semi-autonomous. We acknowledge the hypothetical scenario of the standard being applied to a non-autonomous system but do not pursue this any further.

Moreover, the standard avoids the identification of the *level* of autonomy that an AIS possesses. Often, the *SAE International* levels of autonomy for driverless cars [6] are used to measure the level of autonomy of a given system. There are, however, a variety of frameworks available that define levels of autonomy [7, §3]. Again, by avoiding a prescriptive approach to the level(s) of autonomy to which the standard is applicable, we enable adopters of the standard to choose at what ‘level’ of autonomy necessitates the use of the standard.

Broadly speaking, AIS can be implemented using two categories of artificial intelligence techniques. Symbolic techniques, e.g. logic programming and agent programming languages, use rules and statements to describe the AIS environment and its available actions. Sub-symbolic techniques, e.g. machine learning and neural networks, make inferences from large datasets to build knowledge of the operating environment and how AIS actions will affect it. While it is important to be aware of this implementation distinction, the standard is agnostic in terms of the technology used to achieve autonomy.

C. What is Fail-Safety?

Traditional approaches to designing fail-safe systems rely on understanding failure modes and rates: the latter typically derived from physical properties. The increasing reliability of hardware and the advent of generic, commodity platforms, has re-balanced the focus of design approaches for fail-safe systems to issues of architecture, software failure, and process assurance. These aspects are, however, generally dominated by systematic errors driven by epistemic rather than stochastic uncertainty. Traditionally, tackling these aspects of the design has been approached with techniques such as enhanced process scrutiny, verification, reviews, audits, capability maturity management, validation, and a focus on expert competence. Thus, assuring the fail-safety of software-dependent systems becomes a matter of degrees of confidence and assurance. For AIS we must ensure that:

- it can achieve a known state that prevents or mitigates unacceptable levels of harm in a deterministic way;
- it will do so either: automatically in the event of physical or operational failures, or through external demand driven by a potentially hazardous event or series of events.

The WG has recognised from the beginning that *absolute* safety cannot be achieved: *accidents will happen*. So, the design and development process must be capable of ensuring that the AIS can sustain faults while continuing to provide an

acceptably safe level of operation. This is similar to concepts like risk being “as low as reasonably practicable”⁶ (note, IEEE P7009 is not tied to any one risk acceptability concept). AIS must also be transparent about why and when failures occur to enable end users to validate fail-safe responses. This is related to the concept of Explainable AI [8].

D. Process, Product, or Both?

To be used, a standard has to be both useful and usable. The WG have endeavoured to achieve this for IEEE P7009. The foundational nature of the standard and the lack of application or domain specific context has presented a set of challenges, but also an opportunity to consider the extent to which the standard should address fail-safety by striking a balance between considerations of product and of process.

The IEEE standards development process is rooted in consensus, due process, openness, right to appeal, and balance. It adheres to and supports the principles and requirements of the World Trade Organization’s (WTO) Decision on Principles for the Development of International Standards, Guides and Recommendations. The IEEE (and by extension the WG) operates in active agreement with the WTO principle that standards should not create unnecessary obstacles to trade, and whenever appropriate, should specify requirements in terms of performance rather than design or descriptive characteristics.

A challenge for the WG has been to account for these operating constraints while also recognising the difficulties in achieving quantitative performance requirements of complex systems [9], [10], particularly for software-dependent systems [11]. These intrinsic difficulties cause most current standards to be framed in terms of process rigour as a surrogate for product performance, particularly for non-functional aspects. AIS *are* software-dependent and embody significant, inherently unquantifiable, amounts of uncertainty. Thus, the same or a similar approach offers little realistic prospect of success, when applied to AIS.

The draft standard takes a requirements-driven approach, defining a set of mandatory (*shall*) *Key Requirements* that collectively span both process and product. The *Key Requirements* are complemented by a set of *Critical Properties*, that are required of the AIS. An accompanying mapping between the *Key Requirements* and *Critical Properties* completes the triad of normative elements of the draft standard.

Conformance with the standard is achieved by demonstrating to what degree the system achieves the *Critical Properties* against individual *Key Requirements*, using the mapping. The degree to which a system should achieve the *Critical Properties*, and how to demonstrate this, is left for the adopters of the standard to determine for the particular system and context. The standard thereby avoids being prescriptive, leaving the choice of the acceptable realisation of particular *Key Requirements* to be agreed between the developer (supplier), acquirer, and regulator should there be one.

WG discussions often focused on which *Key Requirements* and *Critical Properties* best expressed the capabilities

⁶<https://www.hse.gov.uk/managing/theory/alarpglance.htm>

an AIS needed to be fail-safe. Many candidate approaches were considered, including immutability, indefeasability, and corrigibility encapsulated in the form of Critical Properties. Ultimately this debate has been resolved in favour of a more conservative *Segregation By Categorisation* approach, which involves both isolation of, and separation of concerns between, fail-safe elements and all other functionality. The resulting approach is analogous to the Trusted Computing Base concept [12], but repurposed for safety. Thus for IEEE P7009, our characterisation of the Trusted Base becomes: “*the small amount of software and hardware that safety depends on and that we distinguish from a much larger amount that can misbehave without affecting safety.*”

Our Critical Properties are: Accountability, Dependability, Interpretability, and Predictability. Collectively, they ensure the delivery of the necessary fail-safety while preventing either endogenous or exogenous misbehaviour of the AIS that would compromise its ability to fail safely.

IV. ONGOING CHALLENGES

This section presents some of the WG’s ongoing challenges. These are core topics for the standard, and the WG is paying close attention to how they are captured to ensure that the standard is useful and usable. A key topic that naturally arises is how to demonstrate the adequate assessment and assurance of the Critical Properties. While this a topic of active AIS research [8], [13], the focus for the IEEE P7009 WG is on ensuring that the draft standard provides adopters with approaches that are not reliant upon trust arguments to support fail-safety. A particular focus is verification and the extent to which either proven or emerging methods and tools [14]–[17] can be usefully applied in the context of the standard. A separate and independent Verification Subgroup has been formed which is addressing some of the following challenges.

A. Verification

As mentioned in §III-D, the WG has endeavoured to avoid adopting conflicting terminology. Here, *verification* involves checking that the system meets its specification; that it is (for example) free of errors, based on representative data, and handles failures safely. In other words, are we building the system correctly, or are we answering the questions correctly. Further, while the standard provides examples of how various properties about AIS might be verified, it will not restrict users to ‘permissible’ verification approaches or tools. Below we present some current issues within the verification subgroup.

What is to be verified? IEEE P7009 is concerned with the existence of, and (autonomous) behaviour within, a system’s fail-safe component; so it focuses on verifying this component, not the V&V of cyber-physical systems in general. Most current standards, such as those identified in §II, are designed for non-autonomous systems. Because we are trying to assess decision-making in new scenarios, the system’s activities may lack detailed requirements. This requires flexible behaviour so that the system can navigate to a safe state that is not necessarily known beforehand. See also §IV-C.

What is different about fail-safety in autonomous systems? AIS bring new risks and challenges for verification [15], e.g. (1) erosion of determinism, and (2) unanticipated, sometimes termed ‘emergent’, behaviours. For (1), current safety approaches often view non-deterministic behaviour as unsafe. When introducing autonomy, we must find a way to reason about and certify safe non-deterministic behaviour. For (2) the AIS might make a decision that it believes to be correct in a given scenario, but this decision might cause a failure (see §IV-B). Moreover, AIS field tests can be very time-consuming and expensive, so it is crucial that the AIS verification process helps to reduce the faults that may hamper field testing.

What are the benefits and drawbacks of techniques that might be used? There are many verification techniques for assessing some requirement of a given system. These include *software testing, simulation-based testing, physical testing, formal verification, runtime verification, and probabilistic verification*. While many of these might be used for the properties and requirements within the draft standard, several verification approaches that work well for other software-based systems may fall short when applied to AIS. Among the verification challenges are: ensuring completeness of ‘correct behaviour’ specifications, multi-directional traceability, proving robustness to adversarial attacks, etc. Several testing coverage metrics have been proposed in the literature, for example neuron, threshold, sign change, value change, sign-sign, and layer coverage, but there is currently little data available on test effectiveness using them. The complexity of the software and the system’s environmental interactions can lead to intractable state-space-explosion, an issue with formal specification. It can also be difficult to build a correct specification of the system’s required behaviour or mission, this can be (for example) because of the complexity of its requirements or because of opaque software components within the system. This is a well known bottleneck with building specifications for cyber-physical systems [18].

However, formal methods that can provide high confidence in their results are useful in the verification of AIS [15]. Also, the concept of corroborative verification [19] where several verification methods that work at different levels of abstraction are applied, in concert to the same AIS could prove useful. Overall, for formal methods to be usefully applied to AIS requires them to be integrated with different verification methods to produce *integrated* techniques, which better cater to their unique verification challenges of [20], [21].

B. Fail-Safe by Design

It is often suggested that, should a system utilise *Autonomous Decision Making* then efforts to provide verifiable fail-safes are doomed. However, there are two important aspects to consider. First, machine-learning systems are not the only way to produce autonomous decision making, there are many other techniques that can achieve this. Perhaps most importantly, modern AIS are rarely monolithic; they have complex and often heterogeneous architectures. Specifically, *modularity* is commonplace across robotics and autonomous

agents. This facilitates architectural approaches that separate effective but hard-to-analyse modules (e.g. machine learning modules) from others, which often use Symbolic AI, for which we can provide stronger guarantees. These heterogeneous architectures support fail-safety through architectural design, where key fail-safe aspects can be captured in particular components. This heterogeneity suggests multiple different verification techniques being applied not only across the system but to individual components [13], [19].

However, as with all autonomous fail-safes, this might move from decision-making behaviour in predictable, bounded scenarios to higher-level, ethical considerations [22]. In many cases this can be avoided, but in unknown and unbounded situations, where safety failures cannot be exhaustively characterised, this will be inevitable [13]. This is discussed in §IV-C.

C. Towards Autonomous Fail-Safety

AIS will fail. Sometimes, a failure does not impact safety, security, or have a societal impact, and a simple STOP procedure can be invoked. In increasingly many cases however, a straight-forward STOP is infeasible or inappropriate. Robots may be engaged in manipulating dangerous items; software components may have complex links into broader digital infrastructures; vehicles may be in hazardous situations; software components may be interacting with users in important ways; immobile systems may create environmental dangers; etc. Here, a straight-forward STOP is rarely the required behaviour. Rather, we need more complex fail-safe components that take us to states that are (intended to be) safe, or at least safer.

Consider a “driverless” car travelling in the overtaking lane (i.e. away from stopping areas) of a highway. A failure occurs and the vehicle must invoke a fail-safe procedure, e.g. to navigate safely to the side of the highway. If the vehicle has a high degree of autonomy (not available at present) the vehicle should autonomously navigate to a safe area and then stop. However, this may depend on the kind of failure that the vehicle has experienced. Here, the fail-safe mode of the autonomous vehicle itself employs autonomous functionality to return to a safe state. Clearly, this better mirrors what a human driver is expected to do by the rules of the road.

It is important that the standard takes such possibilities into account, since we cannot assume that the current low level of systems’ autonomy will persist. Increasing levels of autonomy cause issues for: (a) standard safety-case/fault-analysis techniques, since we cannot foresee all potential hazards; (b) verification techniques, particularly concerning greater uncertainty; and (c) trustworthiness issues, as the step from human to autonomous control will need us to expose and assess the system’s ‘intent’ to retain trust. It is likely that these will require new mechanisms for assuring AIS behaviour [13].

V. EXEMPLAR USE CASES

The WG has found the use of exemplar use cases advantageous, both for developing and for testing the content of the standard. This section highlights two Use Cases that illustrate the types of application and domains in which we

envisage the standard being used. The first use case occurs in a regulated environment; the second considers a scenario in which regulatory oversight is less established, so commercial pressures might be more prevalent.

A. Use Case 1: Tightly Regulated Domain

IEEE P7009 offers guidance on how to safely include autonomous functionality via the provision of fail-safe mechanisms, which is helpful in tightly regulated domains. AIS are evolving at a fast pace and tightly regulated sectors are often not equipped to certify their use.

For example, in the nuclear sector using AIS can protect humans from tasks that expose them to high levels of radiation. A potential application is an autonomous robotic arm with a laser cutting head for cutting (non-fuel) waste material from nuclear operations into smaller pieces for long-term storage. The waste material is secured to a frame during cutting, and the robotic arm is fixed to the operating area floor. Here, the use of autonomy speeds up the planning and execution of a cut, but a human operator may have to approve planned cuts before the system can execute them.

This kind of robotic AIS helps to remove operators from the hazardous area where waste material is being cut, reducing risks related to both the system (collisions, etc.) and the waste material (chemotoxic, radioactive, etc). This system is safety-critical because incorrect targeting of the laser may result in damage to its operating area and cause radiation to leak into surrounding areas. It may also cause damage to itself and require a human to enter the high-risk area to carry out repairs.

In the nuclear sector, regulators require the assembly of a *safety case* to certify that a system is safe for use. However, there is currently little understanding of how AIS such as this one can be correctly captured in a safety case for the nuclear sector [23]. IEEE P7009 can provide regulators with guidance about how these systems can be incorporated into critical scenarios via the provision of effective fail-safe functionalities.

Notably, the inclusion of autonomy improves the safety of a task by removing humans from high-risk scenarios. There is thus a huge benefit to using AIS in nuclear operations, but this is only possible with adequate guidance as defined by standards such as IEEE P7009.

B. Use Case 2: Unregulated or Loosely Regulated Domain

Consider a company operating in sectors where software failures pose a significant concern and many of their products are safety- or mission-critical. There are currently no broad standards for the use of AIS within the sectors that this company operates. Here, IEEE P7009 fills a gap and allows the company to follow a set of guidelines to ensure that their systems are as fail-safe as possible.

Consider an autonomous satellite docking scenario, e.g. [24]. In this domain, national and regional space agencies (e.g. ESA, UKSA, NASA, JAXA, etc) are the *de facto* regulators. However, space is also a new commercial enterprise and satellites can be launched (via private companies) without approval from these ‘regulators’. This increases the possibility

of satellites being launched, running software that has not been designed with fail-safe functionality. Current guidelines in this sector do not explicitly cater for AIS [25].

These systems are inherently mission-critical and software failure can equate to complete mission failure since once deployed, they are not easy to physically access for maintenance/repair. Thus, even though guidelines might be sparse for AIS, it is in the best interest of companies in this domain to incorporate fail-safe functionality as guided by IEEE P7009.

There are many benefits in following IEEE P7009 in an unregulated or loosely regulated domain. Specifically, IEEE P7009 provides guidance for developers about how to ensure the fail-safety of their system. Further, as more detailed regulation/certification processes become available in a given sector at a later date, the evidence gathered by the developer, via IEEE P7009, will provide a basis for this process.

VI. CONCLUSION

After much discussion, the P7009 WG is now focussed on committing settled positions into a draft document. A roadmap guides the way from the working principles identified in §II to a draft standard ready for ballot⁷. This roadmap aligns with the overall architecture of the draft and the set of clauses from which the standard is being assembled. A suite of Scenarios, Use Cases, and Test Cases (the latter in the form of Vignettes) has proven of significant benefit both for testing of developing content but also as a means to assist in the development of draft content. Unfortunately, we are not able to share the contents of an IEEE work-in-progress standard. The other focus for the WG is ensuring that the draft standard not only meets the objectives identified in the PAR but does so in a manner that is useful and usable.

As with many initiatives, particularly those on a scale and challenge such as that presented by the IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems, there is always a risk of the ambition running away from the reality of delivery. Thus far, the IEEE P7009 WG has been able to avoid this outcome. This is testimony to the hard work, good will, and collaborative spirit of the volunteers who have participated and contributed to the WG. Like all previous technological innovations in human history, there will be a balance between the risks and the opportunities that AIS offer. Few can doubt that the potential real-world consequences of failure of AIS will contribute significantly to the risk side of this equation. If the IEEE P7009 standard can offer means to help ensure an acceptable solution to this calculation can be achieved then the WG will have accomplished its mission.

Disclaimer: The contents of this paper do not necessarily reflect the official position of the IEEE, IEEE Standards Association, IEEE P7009 WG, WG Sponsors, or any other organisations with which authors are affiliated either individually or otherwise.

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⁷Balloting: a draft standard is reviewed for approval as an IEEE standard.